

KRISHI ATTRI

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EDUCATION

University of Central Florida

Ph.D., Mechanical Engineering — Post Master's Track

ORCGS Doctoral Fellow | REAL Lab (Advisor: Prof. Hwan Choi)

Orlando, FL, USA

Aug 2026 (Incoming)

Seoul National University

M.S., Mechanical Engineering | GSFS Scholar | Soft Robotics & Bionics Laboratory

Seoul, South Korea

Sept 2024 – July 2026

Villanova University

B.S., Mechanical Engineering | Minor in Mechatronics | Control & Dynamics

Villanova, PA, USA

Aug 2020 – May 2024

EXPERIENCE

Graduate Research Student

Sept 2024 – July 2026

Soft Robotics & Bionics Laboratory, Seoul National University | Seoul, South Korea

- **Government-Funded Humanoid Robot Project:** Contributing to a multi-year government-funded humanoid robot programme at the lab (Phase 1 completed 2024); integrating visuo-tactile SLAM and dexterous in-hand manipulation modules into the Phase 2 full-scale prototype.
- **GaussianFeels — Visuo-Tactile In-Hand SLAM (M.S. Thesis, Advisor: Prof. Yong-Lae Park):** Developed an online visuo-tactile reconstruction and pose-tracking system built around an explicit object-centric 3D Gaussian Splatting map for in-hand manipulation under heavy occlusion. Designed contact-aware Gaussian population management and an occlusion-aware training objective that dynamically reweights RGB, depth, and tactile supervision. Implemented SLAM-loop pose tracking and a frame-0 shape-completion branch (Hunyuan3D-2-mini); evaluated on the FeelSight benchmark.
- **PoP-SLAM — Neural Point Cloud-Based SLAM:** Co-developed a dense visual SLAM system with a projection-first rendering strategy that eliminates nearest-neighbor searches via GPU-vectorised projection of $\sim 15\,000$ neural points/frame. Achieved **best ATE RMSE of 0.75 cm** on TUM-RGBD (outperforming Point-SLAM, NICE-SLAM, ESLAM) at **~ 4 FPS**. Introduced direct occlusion detection via multi-keyframe depth masking.

Robotics & Mechatronics Researcher

Aug 2023 – May 2024

Villanova University | Villanova, PA, USA

- Developed SLAM and visual odometry for quad-wheel robots in GNSS-denied environments via LiDAR-camera fusion, CNN-based feature extraction, and GPU point cloud processing.
- Built a full ROS-based navigation stack (path planning, obstacle avoidance) and configured Arduino/Raspberry Pi microcontrollers for real-time sensor-to-actuator control.
- Implemented probabilistic localisation (2D histogram filter, 1D Kalman filter) and real-time 3D environment mapping via computer vision (edge detection, landmark tracking).

TECHNICAL SKILLS

SLAM & 3D Reconstruction: 3D Gaussian Splatting, Visuo-Tactile SLAM, Neural Point Cloud SLAM, SfM, RGB-D Reconstruction, Object-Centric Gaussian Mapping

Deep Learning & AI: PyTorch, CUDA (custom kernels), TensorFlow, Multimodal Learning, Differentiable Rendering, CNN / RNN / ViT

Robotics & Middleware: ROS, OpenCV, Open3D, Sensor Fusion, Real-time Inference, NVIDIA Omniverse

Hardware & Sensors: DIGIT Tactile Sensor, RGB/RGB-D Cameras, LiDAR, VectorNav IMU, RTK GPS, UR5e, Allegro Hand, Vicon/OptiTrack

Programming: Python, C/C++, CUDA, MATLAB, Arduino, $\LaTeX$, HTML

Tools & Systems: Linux (Ubuntu), SOLIDWORKS, Blender, Git/GitHub, KiCAD, Jupyter Notebooks

PROJECTS & ACHIEVEMENTS

Computer Vision Object Detection App | Python, PyTorch, React, Flask

Summer 2024

- Built a full-stack web application for real-time image and video object detection. Implemented and benchmarked five detection models (Faster R-CNN, Mask R-CNN, RetinaNet, Keypoint R-CNN, SSDlite) using PyTorch and TensorFlow with COCO dataset evaluation. Evolved through 8 versions from single-model photo processing to multi-model video comparison.

Capstone: Plant Lifting Device for 3D Imaging | FMC Corporation (Sponsored)

Aug 2023 – May 2024

- Led multidisciplinary team through full concept-to-prototype lifecycle (actuators, motion systems, wiring, waterproofing); awarded **1st Place, Most Innovative Solution** at the university capstone showcase.

CERTIFICATIONS & AWARDS

NVIDIA Computer Vision Nanodegree (Udacity, 2024) · **K-MOOC Soft Robotics & Applications** (SNU, 2024) · **Robotics Specialisation** (Coursera/UPenn, 2021)

GRE: Quant 166 (80th pctl), Verbal 153 · **Dean's List:** Fall 2020, Spring 2021 · **Capstone 1st Place** (2024) · **ICE Competition 3rd Place** (2020)